

13

GROWTH MODELING USING THE DIFFERENTIAL FORM: TRANSLATIONS FROM STUDY OF FISH GROWTH¹

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Growth Modeling using the Differential Form: Translations from Study of Fish Growth

Theories of development all promote the idea that human functioning proceeds simultaneously on multiple levels (e.g., cells to society) and time-scales (e.g., seconds to decades). Meta-theoretical frameworks (Baltes, Lindenberger, & Staudinger, 2006; Bronfenbrenner & Morris, 2006; Ford & Lerner, 1992; Gottlieb, 2007, Overton & Lerner, 2012) highlight the importance of understanding how, when, and why individuals change (see also Li, 2003). Domain specific theories too suggest how, when, and why particular attributes, abilities, and characteristics change with time and/or age and in relation to specific biological, psychological, social, and environmental factors. To deal with the complexities involved in studying developmental processes, methodologists have advanced a wide variety of longitudinal methods for accurately charting how individuals and, more generally, “entities” (since we will be looking at models of fish growth) change across time. Innovations in longitudinal study design continually push forward researchers’ ability to discover and explain *intraindividual change* processes and the *interindividual differences* therein (see McArdle & Nesselroade, 2014).

Cohort sequential, longitudinal panel, and accelerated longitudinal studies – wherein multiple reports are obtained from multiple individuals at regular intervals – have enhanced our capability to describe and test hypotheses about within-person change (Hertzog & Nesselroade, 2003; McArdle & Anderson, 1990; McArdle & Bell, 2000). Data from these types of studies are typically analyzed using some variant of *growth modeling* (McArdle & Nesselroade, 2003). Certainly, and in large part following McArdle’s innovations, growth modeling has emerged as the primary method for examination of developmental processes. McArdle’s extensions for modeling of multivariate change and latent change

(differences or differentials) are at the forefront of current possibility. Celebrating Jack McArdle’s continual push for better and more robust models, we use this opportunity to explore recent and classic ecological literature on growth modeling and develop some nascent ideas about how the models in use there may be applied in study of human development. In the sections that follow, we review the benefits of considering growth models as *change process models* and explore the mathematical and theoretical connections between growth models used in ecology for study of fish growth and models of development in humans. We finish the chapter with some notes on implementation and some suggestions for what the future may bring.

Reformulating Growth Models in the Differential Form

In the past few years we have attempted to organize our work on growth modeling with respect to the goals of longitudinal research, issues surrounding practical implementation in both SEM and MLM frameworks dealing with nonlinearity (both with respect to time and to the model parameters and random coefficients), and how various models are used to examine qualitative and quantitative interindividual differences and intraindividual changes in features of growth trajectories (e.g., growth mixture and/or age-conditioned growth models). Along the way we are continually wrestling with how growth models and, more generally, longitudinal models provide an interface for reconciling theory and data. McArdle pushed us to consider the theoretical model of change – and, in particular, forwarded the latent change score framework that allows for more precise articulation of theory and representation of longitudinal data. Our current exploration follows hints we found on an *n*th reading of McArdle’s classic chapter in the *Handbook of Multivariate Experimental Psychology* (McArdle, 1988).

In brief, growth models are used to describe individuals’ longitudinal trajectories and the between-person differences in those trajectories. Specific mathematical functions (e.g., linear, quadratic, exponential, sigmoid, sinusoidal) are used to describe the observed *output* of a latent change process. For example, the linear growth model describes the observed outputs, y_{it} , as a linear function of time,

$$y_{it} = \beta_{0i} + \beta_{1i}(\text{time}_{it}) + r_{it} \quad (13.1)$$

where β_{0i} is a person-specific *intercept* (predicted value of y) located at $\text{time} = 0$, β_{1i} is a person-specific *slope* that is interpreted as the rate of linear change in y for a one unit change in time , and r_{it} are time-specific residuals. Pushing for more explicit rendering of the *change process* that is producing the observed output, McArdle and Hamagami (2001) presented a formulation of the standard latent growth model that is based on successive changes. In general form, this (univariate) *change score model* can be written as

$$y_{it} = [\beta_{0i}] + \left(\sum_{t=2}^t \Delta y_{it} \right) + r_{it} \quad (13.2)$$

where individual i 's score at occasion t , y_{it} , is the sum of an initial score, β_{0i} (interpreted as an intercept or baseline score) and all of the subsequent changes that have accrued up to that time (sum of Δy_{it} from $t = 2$ to t). This (re-)rendering of the basic growth model makes the underlying change process explicit. Specifically, we see that, to obtain equivalence between Equations 13.1 and 13.2 (i.e., the standard linear growth model and the latent change rendition of the model), the model for all the successive changes must be

$$\frac{\Delta y_i}{\Delta t} = \beta_{1i} \quad (13.3)$$

where β_{1i} is a person-specific rate of change. This *difference equation*, then, is an explicit rendering of the change process underlying the linear growth model – and serves as the *theory of change* that can be tested against repeated measures data.

Portending the presentation of models for fish growth that will come later, we highlight that the contrast McArdle makes between the models used to describe *outputs* (traditional regression-based model for y_{it}) and models used to describe the *change process* (difference equation model for Δy_{it}) has also emerged in recent discussions of fish growth (Enberg, Jørgensen, Dunlop, Varpe, Boukal, Baulie, Eliassen, & Heino, 2012). In particular, a contrast is made between fish *size*, which characterizes an individual's state, and fish *growth*, which is a process that leads to that state. Models of the growth process are formulated as difference or differential equations and derived from bioenergetic and behavioral principles, and the models of size measurements (e.g., length, weight, mass) are integral solutions of those change process equations.

When presenting the change score models, McArdle and colleagues have mostly been working with repeated measures data with between 2 and 20 measurement occasions obtained at monthly, yearly, or longer intervals – data from longitudinal panel studies. Representations of the *time* variable in these applications are usually *discrete*, in that the number of measurements is conceived as finite. The discrete representation of time has been particularly useful in SEM implementations, where the measurements for a given time period can be invoked as a unique factor loading and/or variable. Outside the SEM framework, representations of the time variable are often continuous (e.g., chronological age), in that the number of (potential) measurements is conceived as infinite. Continuous representations of time have been particularly useful, because they explicitly invoke notions of organismic continuity that sit at the core of many developmental and process-oriented theories. Although mostly a semantic nuance in practice, continuous time models support straightforward description of changes over spans of time while discrete time models support description of changes between moments in time (see Ram & Reeves, in press). In practice, growth modelers are fluidly moving between discrete and continuous time with ease – invoking discrete or continuous interpretations of the time variables wherever convenient. For example, the change process described by Equation 13.3, which is a discrete time

difference equation where the time variable spans a sequence of integer values, $t = 1, 2, 3, \dots T$, is easily rewritten as a continuous time *differential equation* where the time variable spans the entire real number line (see e.g., Boker, 2001),

$$\frac{dy}{dt} = \beta_{1i} \quad (13.4)$$

Once made explicit as a difference or differential equation, the simplicity of this particular theory of change is clear. We see clearly that the linear growth model implies that the intraindividual change processes can be entirely described by a single number, β_{1i} , that does not vary over time, is not related to where an individual is in the change process (e.g., level of y_{it}), and is not related to any aspect of context. As such, the linear growth model seems at odds with key tenets of most developmental theories. The re-rendering of the model as a difference or differential equation makes this explicit – and thus brings us closer to the actual phenomena of interest: *change processes* (see McArdle, 1988, 2009).

More generally, the explicit descriptions of the underlying change process afforded by difference and differential equations prompt consideration and testing of more precise, appropriate, and complex hypotheses about behavioral change. Most theories of human behavior suggest that people are dynamic, and that the change processes that drive their development are probably not constant. A few simple extensions can illustrate how working directly with the differential equation models facilitates articulation and construction of more complex and realistic theories of change. For example, we may theorize that there is endogenous dependence in the growth process. That is, the progression of the process depends on the level of output or ability already achieved,

$$\frac{dy}{dt} = ky_t \quad (13.5)$$

Also, we can incorporate the idea that growth is bounded by endogenous or exogenous constraints. Incorporating an upper bound or *carrying capacity*, Y_∞ , into Equation 13.5, we get the change process model that will be the focus of our discussion through the rest of the chapter,

$$\frac{dy}{dt} = k(Y_\infty - y_t). \quad (13.6)$$

This model indicates that the closer y_t is to the carrying capacity, the smaller the rate of change. The general idea is that translating theoretical propositions into differential equations is much more straightforward than working directly with the analytical equations that are actually fit to the data. The “lingua-franca” of differential equations provides for straightforward articulation of knowledge about change processes, highlights the potential inadequacy of currently popular models (e.g., linear growth), and opens up a wide range of possibilities for articulating the complexities of behavioral change.

Translating Models from Ecology to Developmental Science

A wide variety of growth models are being used to study change processes in many fields, including agronomy, biology, chemistry, ecology, economics, engineering, forestry, ichthyology, medicine, psychology, sociology, and zoology. Indeed, general books on growth modeling are filled with long lists of potential models – differential equations that describe many types of change processes. For instance, Panik (2014) lists 20 or so “mainstream” models for change (linear, logistic, Gompertz, Weibull, von Bertalanffy, Chapman-Richards, Janoshek, Morgan-Mercer-Flodin, etc.). As far as we know, only a handful of these models have made their way into the developmental and psychological literature (see reviews in Preacher & Hancock, 2015; Wood, Steinley, & Jackson, 2015). Our overall view is that uptake might be hastened when these models are recast in ways that facilitate mapping to theoretical models of change. McArdle (1988) has long suggested that this is where the *differential equation* formulations can be quite useful.

In our exploration of how differential equation models are used to examine change, we noted (as others have) that naturally occurring processes that underlie many self-organizing phenomena, including pattern formation and self-regulation, may be described using relatively simple *activator+inhibitor* models (Ram, Gatzke-Kopp, Gerstorf, Coccia, Morack, & Molenaar, 2015). In brief, these two-component systems consist of an activator that stimulates its own production as well as the production of the inhibitor, which in turn represses production of the activator (e.g., Geier & Meinhardt, 1972). The utility of the models is particularly evident in systems biology, where similar two-component “competing processes” models afford description of a wide variety of qualitative and quantitative interindividual differences in change processes (Murray, 2003; Turing, 1952). In our search for recent innovations in growth modeling, we delved into the ecology literature and found work (see special issue of *Fisheries Research* on growth; Maunder, Crone, Punt, Valero, & Semmens, 2016) that, when consumed alongside McArdle’s (1988) treatise, inspired us to consider further how models used in biology may be useful in our study of human development and behavioral change. Here, we draw on a model that is at the core of many fishery applications – the von Bertalanffy (1956) model. Similar to the *activator+inhibitor* model, this *anabolism–catabolism* model also affords theoretical and empirical description of a wide variety of qualitative and quantitative interindividual differences in intraindividual change.

In the sections that follow, we approach the model in three ways. First, we present the familiar integrated mathematical form used to describe observed repeated measures data – traditional regression-based model of *outputs* (i.e., size). Plots of prototypical trajectories illustrate how the parameters of the model map to specific features of the growth trajectory. Second, we outline how theoretical propositions about the processes driving intraindividual change are formalized in the differential form – differential equation model of the *change process*. Simulations are used to illustrate the flexibility in development afforded by the

anabolism–catabolism “competing processes” model. Finally, we highlight some of the ways the von Bertalanffy growth model is being adapted and extended to accommodate plasticity in growth, interindividual (or interspecies) differences and the role specific biological and environmental factors play in growth. In doing so we attempt to discern how innovations forwarded in study of fisheries (and other animal and plant populations) may be applied in the study of human development.

A von Bertalanffy Growth Model for Fish Growth: The Integral Form

The von Bertalanffy growth function is widely used in study and management of fish populations (Quinn & Deriso, 1999). The model is often written as

$$L_t = L_\infty (1 - e^{-K(t-t_0)}) \quad (13.7)$$

where L_t , the length of the fish at time t , is modeled as a nonlinear function of the asymptotic length L_∞ , the growth rate K , and the time (or age) t_0 at which the fish has a length of zero. A representative trajectory is shown in Figure 13.1, with the parameterized features of the trajectory highlighted: $L_\infty = 100$, $K = 0.046$,

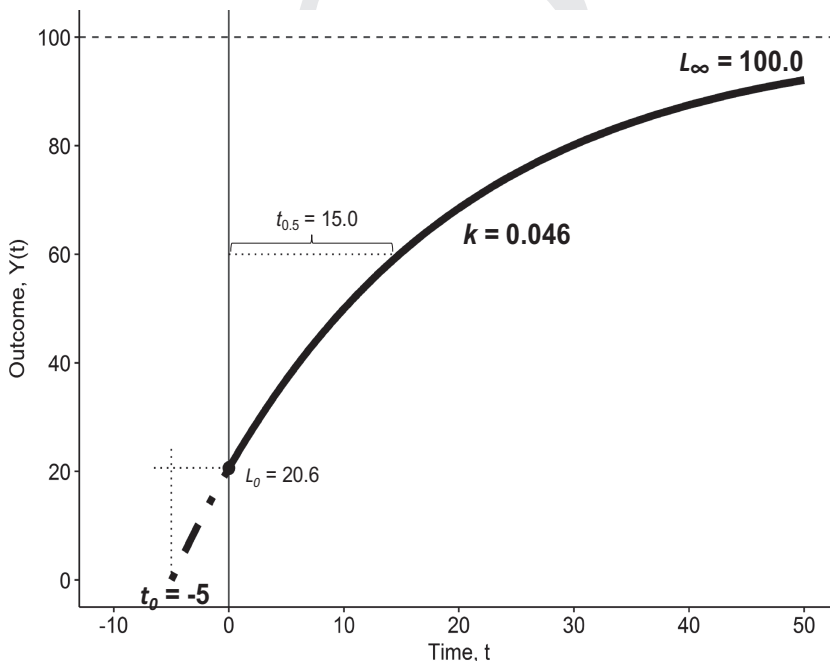


FIGURE 13.1 Illustration of the von Bertalanffy growth model. Trajectory for L_t , the length of a fish at time t , is modeled as a nonlinear function of asymptotic length L_∞ , growth rate K , and the time (or age) t_0 at which the fish has a length of zero.

and $t_0 = -5.0$. An *equivalent* specification that trades t_0 (the constant indicating the timing of $L_t = 0$) for an L_0 intercept (indicating expected length at $t = 0$),

$$L_t = L_0 e^{-Kt} + L_\infty (1 - e^{-Kt}), \quad (13.8)$$

may be useful in some empirical situations. The trajectory in Figure 13.1 provides the classic depiction of growth in fish length, an asymptotic curve progressing from birth length of $L_0 = 20.62$ with continually declining growth rate toward an asymptote. As may be noted mathematically or visually, this 3-parameter von Bertalanffy model is functionally equivalent to the negative exponential model used by McArdle to describe age-related change across the life span in individuals' intellectual ability (e.g., McArdle, Grimm, Hamagrami, Bowles, & Meredith, 2009), and is similar to many other nonlinear trajectory models, including the Logistic, Gompertz, Richards and Schnute models that the McArdle group has used in recent years. In-depth discussion of the connections amongst all of these models can be found in Banks (1994) and Panik (2014).

For examination of interindividual differences in intraindividual change, the model is easily cast as a nonlinear growth model in either the multilevel or SEM framework and used to simulate and/or describe individual trajectories (Grimm, Ram, & Estabrook, 2017; McArdle & Nesselroade, 2014). For example, expanding the model to accommodate interindividual differences in all three parameters and presence of measurement/residual error, we can generate a collection of individual trajectories using the multilevel model

$$L_{it} = L_{\infty i} (1 - e^{-K_i(t-t_{0i})}) + r_{it} \quad (13.9)$$

where L_{it} , the observed outcome for fish i at time t , is modeled as a function of fish-specific coefficients for the upper asymptote, growth rate, and horizontal shift features, $L_{\infty i}$, K_i , and t_{0i} , which are in turn modeled as

$$L_{\infty i} = \pi_{00} + \dots + u_{0i} \quad (13.10)$$

$$K_i = \pi_{10} + \dots + u_{1i} \quad (13.11)$$

$$t_{0i} = \pi_{20} + \dots + u_{2i} \quad (13.12)$$

where the π s are sample-level parameters that describe the trajectory of the “prototypical fish”, and the u s describe between-fish differences in the features of the trajectories. Interindividual differences (that are assumed multivariate normally distributed) in the parameterized features are shown in the panels of Figure 13.2. The top left panel, for instance, isolates differences in the upper asymptote ($L_{\infty i}$). Similarly, the middle left panel isolates differences in growth rate (K_i), and the lower left panel isolates initial length or “launch time” (L_{0i} or t_{0i}).

Setting up a later point of discussion, we note here how the growth rate parameter may be transformed for additional interpretive value. The rate parameter K_i has units of reciprocal time. Transformation of K_i into a *doubling time* or *half-life*

284 Nilam Ram & Xiao Yang

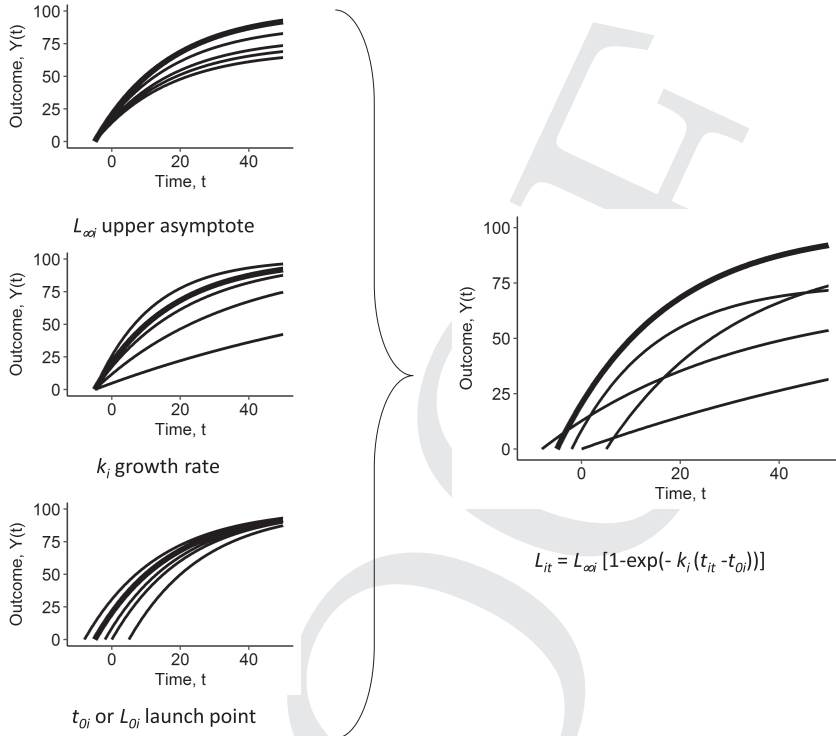


FIGURE 13.2 Depiction of interindividual differences in the trajectory features parameterized in the integral form of the von Bertalanffy growth model. The top left panel isolates differences in the upper asymptote ($L_{\infty i}$), the middle left panel isolates differences in growth rate (K_i), and the lower left panel isolates initial length or “launch time” (L_{0i} or t_{0i}). The right panel collects the differences in all three parameters to illustrate the range of potential interindividual differences in intraindividual change captured by the model.

parameter that is scaled in units of time facilitates discussion of the characteristic time-scale of the growth process, and the interindividual differences therein. Specifically, we use the equation

$$t_x = \frac{1}{K} \ln \left(\frac{L_{\infty} - L_0}{L_{\infty} (1 - x)} \right) \quad (13.13)$$

to calculate the half-life for each individual as $t_{0.5i} = \frac{1}{K_i} \ln 2$. For example, the time it takes to travel half the distance between $L_0 = 20.6$ and $L_{\infty} = 100$ in Figure 13.1 is calculated from $K = 0.046 \text{ years}^{-1}$ as $t_{0.5} = 15$ years. The same transformation can also be used to calculate a so-called *longevity estimate* that is conceptualized as the time it takes to reach a specific fraction x of L_{∞} . Fish classics (Fabens, 1965; Ricker, 1979) suggest interpretation of $t_{0.95}$ or $t_{0.99}$ (see Callet, Smith, Mollet, & Goldman, 2006).

Bringing the three sets of coefficients together, the right panel of Figure 13.2 provides an overview of the range of interindividual differences in intraindividual change covered by this model. As with other nonlinear models, the von Bertalanffy model expands the possibilities for describing meaningful heterogeneity far beyond what can be captured by the linear growth model. Although not written out explicitly here, explanatory predictors can be added to explain both the growth process and the interindividual differences therein (see Ghisletta, Cantoni, & Jacot, 2015). We will touch on some of those extensions later. The main message we want to convey at this juncture is that the von Bertalanffy growth model is similar in structure and layout to other growth models being used in study of development (McArdle & Nesselroade, 2003; Ram & Grimm, 2015).

Bioenergetic Determinants of Growth: The Differential Form

In our attempt to better understand the ways in which theoretical models can be mapped to difference and differential equation models of change, the von Bertalanffy growth model has been particularly useful, in part because it was formulated from a very general bioenergetics model – i.e., from basic theoretical propositions. Both the initial development of the model, and the more recent extensions, are attempts to articulate specific physiological phenomena in a mathematical terminology that provides precise mapping of theoretical concepts to model parameters and longitudinal data.

In brief, individual organisms are considered analogous to reacting chemical systems that obey the laws of mass action (i.e., that the rate of a given chemical reaction is proportional to the activities or concentration of the reactants). The physiological processes responsible for the changes in mass of an organism are classified into two broad categories, *anabolism* (e.g., the synthesis or building up of body materials and the structures that support ability) and *catabolism* (e.g., the breaking down or maintenance of body materials and structures supporting ability). The basic von Bertalanffy growth model is then developed from the idea that growth is the net result of the antagonistic interplay of *anabolism* and *catabolism*. Simplistically,

$$\text{growth} = \text{anabolism} - \text{catabolism}$$

or restated in bioenergetic terms, $\text{growth} = \text{energy assimilation} - \text{energy loss}$. In line with basic tenets of lifespan development, the model explicitly acknowledges that during any given span of time, there is *both* gain and loss (Baltes et al., 2006).

Formalizing the propositions into mathematical form, the change process is described by the differential equation

$$\frac{dW}{dt} = HW^d - mW^n \quad (13.14)$$

where H and m are coefficients of anabolism and catabolism respectively, and the exponents d and n indicate how the two processes scale with organism size, W .

Conceptually, the exponent for anabolism, d , may be thought of as the proportion of the organism that can convert inputs into energy (i.e., “effective resorbing surface”; how much of the organism’s body material can be restructured). In parallel, the catabolism exponent, n , may be thought of as the proportion of the organism that uses energy (i.e., is engaged in metabolism). In fish populations, for instance, early empirical evidence suggested that the acquisition of resources, anabolism, is proportional to body surface and thus scales with $d = 2/3$. In contrast, the catabolic costs of maintenance (e.g., basal metabolism and reproductive investment) are assumed directly proportional to body mass and thus scale with $n = 1$. The differential scaling of anabolism and catabolism suggest that as the fish grows larger, more and more of the available energy will be used for maintenance, so that growth slows down and eventually stops – as depicted by the nonlinear trajectory in Figure 13.1.

The size-dependent change in the rate of growth has parallels in human development. Indeed, many lifespan developmental theories highlight the importance of ontogenetic changes in the relative allocation of resources to growth versus maintenance (Baltes et al., 2006). For instance, socioemotional selectivity theory (Carstensen, Isaacowitz, & Charles, 1999) suggests that as people age their motivational priorities gradually shift from information seeking (analogous to anabolism) toward emotion regulation and maintenance (analogous to catabolism). Similarly, motivational theory of lifespan development (Heckhausen, Wrosch, & Schulz, 2010) suggests that personal growth is facilitated through interplay of primary control processes directed at changing the world to bring the environment into line with one’s wishes (enacting goals) and secondary control processes directed at changing the self to bring oneself into line with environmental forces (e.g., giving up goals). Trajectories of successful aging thus might also be explicitly modeled with respect to individuals’ selective allocation of resources to primary control and secondary control processes using a von Bertalanffy model. Individual differences in allocations to each category of processes (e.g., differences in H and m) or contextual differences in opportunity to exert control (e.g., sociodemographic differences in d) provide for heterogeneity in individuals’ life course trajectories. Growth in intellectual ability may also be simultaneously driven by anabolic and catabolic processes. For example, changes in ability may be the outcome of interplay between neurogenesis and neural pruning (Stiles & Jernigan, 2010). In sum, the “competing processes” model invoked by the allometric relations in Equation 13.14 provides a general framework for articulating a variety of developmental theories.

Noting the parallels between bioenergetic principles and propositions forwarded in lifespan developmental theory, we explored further how differences in the coefficients and exponents in the change process model in Equation 13.14 may invoke differential development. To do so we built a simulation engine that allowed us to generate synthetic data for a range of coefficient values. Specifically, we simulated data from the differential equation model across many parameter combinations (varying H , m , and d , and assuming $n = 1$) using a feed-forward

procedure to track and plot how both the change process (growth, dW/dt) and the outcome of the change process (size, W) evolved over time. Data were generated iteratively from $t = 0$ to 50 with a step-size of $dt = 1.0$, starting with an initial value for $W_i = 0.01$. A select set of the resulting change process data, for parameter combinations where $H \in [0.75, 0.85, 0.95]$, $m \in [0.25, 0.35, 0.45]$ and $d \in [0.62, 0.67, 0.72]$, are shown in Figure 13.3.

As seen throughout the array the *growth rate* curves (dW/dt on the y -axis and *time* on the x -axis) generated from the von Bertalanffy model are generally

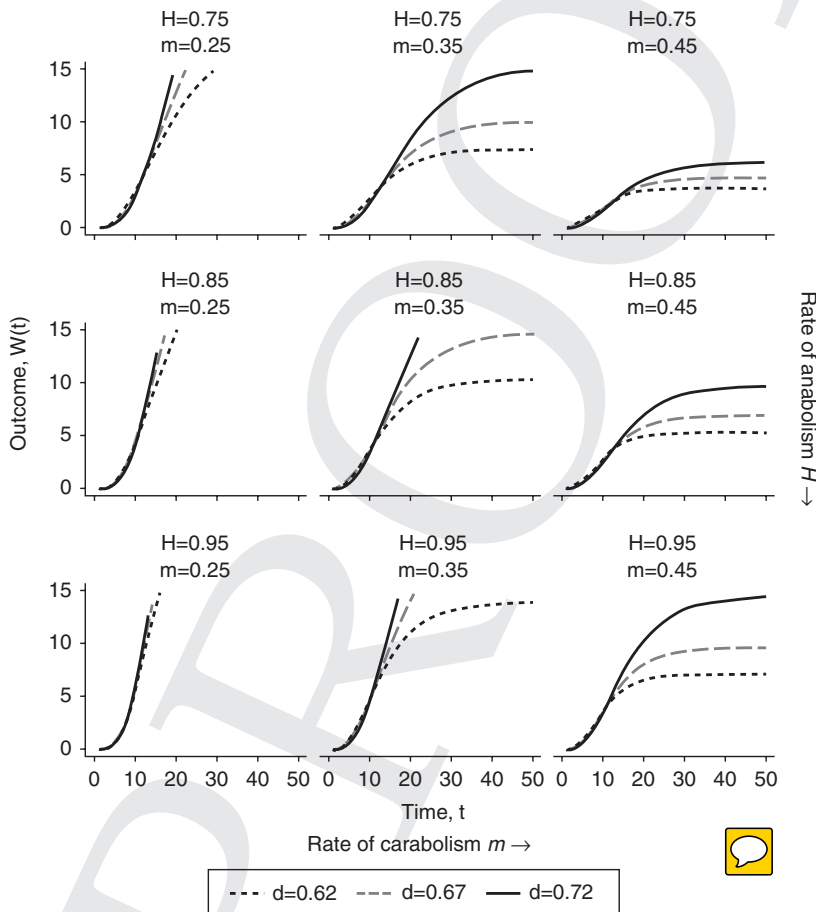


FIGURE 13.3 Simulation-based depiction of how different values of the allometric coefficients and scaling exponents in the differential form of the von Bertalanffy growth model produce different growth rate (dW/dt) curves. As the rate of anabolism (H) increases across panels, top to bottom of the figure, the asymptotes are higher. As rate of catabolism (m) increases across the panels, left to right, the asymptotes are reached earlier. Within a given panel, when the anabolic scaling factor is higher (high d), the asymptotes are higher and reached later.

characterized by an inverted-U shape where the growth rate increases and then decreases over time. Patterns across the grid are instructive about how the allometric relations surrounding anabolism and catabolism influence the growth process. Starting with how anabolism scales with mass, we see that larger values of d produce larger and longer lasting growth. For example, looking within any specific panel, we see that the peak of the blue line (high $d = .72$) is both higher and arrives later than the peak for the red line (low $d = .62$). Indeed, it makes sense that when a greater proportion of mass is available as an effective resorbing surface, more of the assimilated energy is available for growth. The anabolic processes are able to contribute more to growth before they are “overtaken” or “overwhelmed” by the catabolic processes. Similarly, looking across panels vertically, we see that higher values of H are also associated with larger humps. When anabolic processes are being directed toward growth in an “efficient” way (large H), growth rates are generally larger. In contrast, at low values of H , when anabolic processes are not being directed toward growth, growth is bounded or never really gets started. Looking across panels horizontally we see that higher values of m are associated with lower growth rates generally, and smaller humps. Indeed, when the rate of metabolism is high (large m), resources are directed toward maintenance rather than growth. In contrast, when the rate at which body materials or structures are being broken down is low, growth rates are high for a longer period of time. Altogether, the collection of simulated trajectories illustrates the flexibility in growth patterns afforded by a “competing processes” model like the von Bertalanffy model, and how the heterogeneity in trajectories is tethered to systematic variation in the coefficients and scaling factors describing the allometric relations between specific physiological processes (anabolism and catabolism) and growth.

Connecting the Differential and Integral Forms

In Figure 13.3 we examined how variation in the allometric coefficients and scaling exponents in Equation 13.14 produced a variety of growth rate curves. The illustration demonstrates individual differences in the *growth process* (i.e., changes) produced by the differential form. Our general push has been toward use of the differential equation model – same as McArdle’s push for use of the change score model. The thinking is that proximity to the change process model facilitates translation of theoretical propositions into a parameter space. We did, however, do something similar in Figure 13.2. There, we highlighted how the model provides for description of interindividual differences in the three *features* of the trajectories (upper asymptote, shape, launch point) parameterized in Equation 13.7. For continuity, the array of *size* trajectories that are produced by the array growth processes displayed in Figure 13.3 are shown in Figure 13.4. Thus, one can see visually how the specific patterns of growth manifest in the outputs. Looking vertically across the panels in Figure 13.4, as the rate of anabolism (H) increases, the asymptotes are higher. Looking horizontally across panels, as rate of catabolism (m)

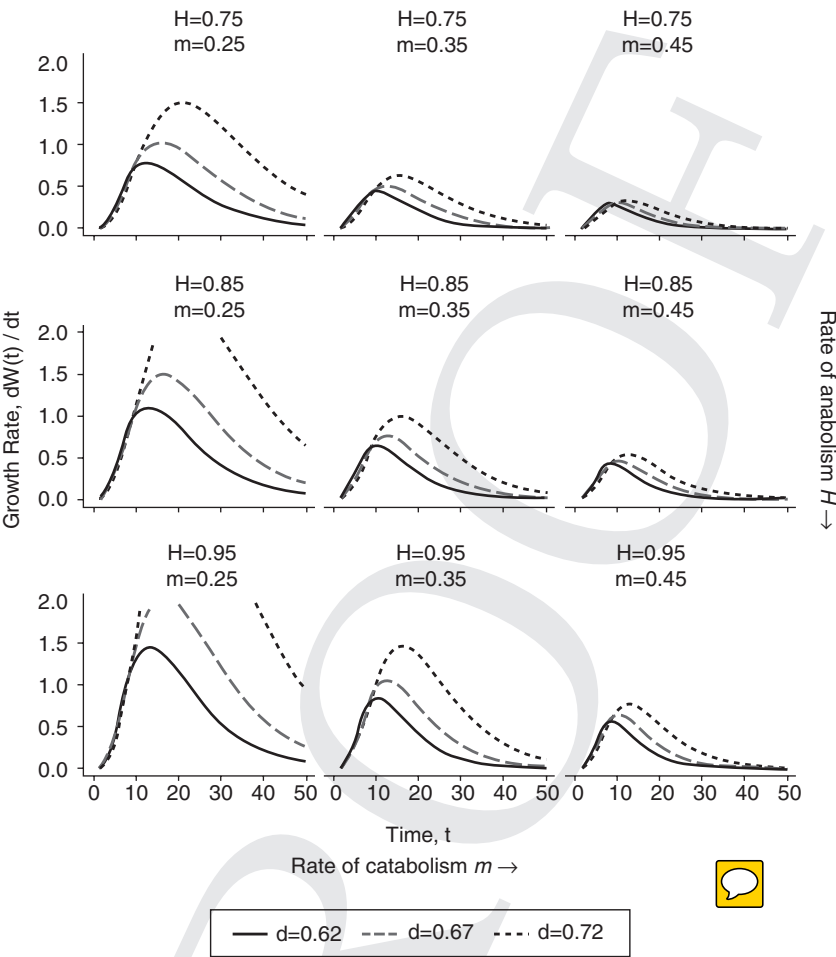


FIGURE 13.4 Simulation-based depiction of how different values of the allometric coefficients and scaling exponents in the differential form of the von Bertalanffy growth model produce different trajectories for size (W_t).

increases, the lower asymptotes are reached earlier. Looking within a panel, when the anabolic scaling factor is higher (blue line = high d) the asymptotes are higher and reached later. Most notable is that the asymptote changes in relation to all three parameters being varied. Building from the visual illustration, the mathematical relations illustrate more formally how the parameters (H , m , d , n) in the differential equation model (which map to theoretical principles) are linked to the three parameters (Y_∞ , K , and t_0) in the integral equation model (which map to visual features of the size trajectories).

von Bertalanffy used three assumptions to obtain the model for length, Equation 13.7, from the differential model in Equation 13.14 (see van Poorten,

290 Nilam Ram & Xiao Yang

2016). Notably, although these assumptions have both a theoretical and empirical basis, they also happen to be mathematically convenient. The bioenergetics model for growth is formulated specifically in units of mass (i.e., weight, W),

$$\frac{dW}{dt} = HW^d - mW^n \quad (13.15)$$

First, it is assumed that catabolism varies directly with mass, $n = 1$. Setting $dW/dt = 0$ and $n = 1$, Equation 13.15 can then be rearranged to solve for asymptotic weight as

$$W_\infty = \lim_{t \rightarrow \infty} W_t = \left(\frac{H}{m} \right)^{\frac{1}{1-d}} \quad (13.16)$$

and integrated to obtain the generalized equation for weight

$$W_t = W_\infty \left(1 - e^{-K(t-t_0)} \right)^{\frac{1}{1-d}} \quad (13.17)$$

where

$$K = m(1 - d). \quad (13.18)$$

Second, it is assumed that anabolism varies as the $2/3$ power of mass, $d = 2/3$. Thus, the model simplifies to

$$W_t = W_\infty \left(1 - e^{-K(t-t_0)} \right)^3. \quad (13.19)$$

Of note, and a point we will come back to later, the cubic exponent in the model for mass indicates that this model exists in a 3-dimensional world where fish have length, width, and height. To obtain the specialized 1-dimensional model for length, the third assumption invokes what we would call a *measurement model* that links weight (W) and length (L). In particular, the relation is assumed as

$$W = aL^b \quad (13.20)$$

with $b = 3$ (for fish exhibiting isometric growth). Applying the assumptions that $n = 1$, $d = 2/3$, and $b = 3$ (and $a = 1$), the equations simplify to the model used for Figures 13.1 and 13.2,

$$L_t = L_\infty \left(1 - e^{-K(t-t_0)} \right). \quad (13.21)$$

Three important implications for study of individual differences in growth become apparent when examining the relations between the parameters of the differential equation model for change and the parameters of the integrated model for length: (13.1) The growth parameter K is proportional to the catabolic parameter, $K = m/3$ (from Equation 13.18). This means that, in the 3-parameter von Bertalanffy model, the rate at which an individual moves toward the asymptote is *only* related to rate of catabolism, and not to anabolism. Thus, the external

factors related to interindividual differences in K_i may be interpreted as factors that influence catabolic processes. (13.2) The asymptote L_∞ , is a scaled ratio of mass-specific consumption and metabolic costs (H and m)

$$L_\infty = \left(\frac{H}{m} \right) a^{-\frac{1}{3}}. \quad (13.22)$$

That is, the asymptotic size is related to *both* anabolism and catabolism. Thus, external factors related to interindividual differences in $L_{\infty i}$ may be interpreted as factors that influence anabolic *and/or* catabolic processes. Lack of association with $L_{\infty i}$, however, should not be interpreted as lack of influence because factors that influence both consumption and metabolic costs may cancel out. For example, temperature exponentially increases both fishes’ basal metabolism and their feeding rate. The (lack of) relation between temperature and asymptotic length misses the influence on the underlying metabolic processes driving growth. (13.3) Because K and L_∞ are both directly related to m , interindividual differences in the rate of growth and asymptote will be correlated. As such, it is very likely that external factors that are related to $L_{\infty i}$ will also be related to K_i . Thus, it is the differential relations that may be particularly useful for parsing which factors act on which underlying processes. In summary, when selecting and examining predictors of interindividual differences in growth (i.e., in Equations 13.10 to 13.12), researchers should carefully consider how each feature of the trajectory is related to the underlying (anabolic and catabolic) processes driving intraindividual change.

Example Extensions

The von Bertalanffy growth model is, of course, too simple and based on questionable assumptions. Indeed, ecologists are continually discussing and developing extensions that might accommodate additional biological considerations (e.g., investment in reproductive tissues) and the wide variety of (fish) population demographic and environmental factors known to influence growth (e.g., age-structure and density of population, spatial variation in availability of prey, behavioral changes induced by predators, temperature, oxygen concentration). With an interest in how the modeling extensions are conceived and constructed in ecology, we highlight two variants of the von Bertalanffy growth model that have emerged to accommodate variation in growth related to biological investment in reproduction and seasonal changes in environment.

Biphasic growth. A pervasive concern in the application of the von Bertalanffy growth model is that the model often describes growth of adult fish much better than growth of juveniles (Jones, 2002). Extrapolation to younger or older ages is often unrealistic. The root cause of these concerns is the assumption that the allometries of anabolism and catabolism are invariant over age. Contrary to the assumption, there are ontogenetic shifts in diet, activity, and energy allocation.

292 Nilam Ram & Xiao Yang

As fish grow, so too does cruising speed, size of spatial range, attack rate, gape size, and visual acuity – all of which have important implications for consumption and metabolism, and thus growth. Of particular importance in the management of fisheries, for example, is the increased allocation of resources to reproductive tissue and behavior around the time of maturation.

A variety of models have emerged to account for reproductive investment. Generally, these models accommodate the possibility that one or more parameters shift over time. For example, the typical von Bertalanffy model can be supplemented with a time-varying factor A_t that modifies the growth rate as a function of time (Soriano et al., 1992). The model becomes

$$L_t = L_\infty [1 - e^{-KA_t(t-t_0)}] \quad (13.23)$$

where the function given by

$$A_t = 1 - \frac{h}{(t - t_h)^2 + 1} \quad (13.24)$$

introduces two new parameters: t_h defines the time (i.e., age) at which the transition between the two growth phases occurs, and h determines the magnitude of the maximum difference between the functions for Phase 1 and Phase 2. Depictions of a few trajectories are shown in Figure 13.5A, with values of the t_h and h parameters specifically chosen to highlight possibility of identifying differences in timing of the phase-switch and the extent of the switch.

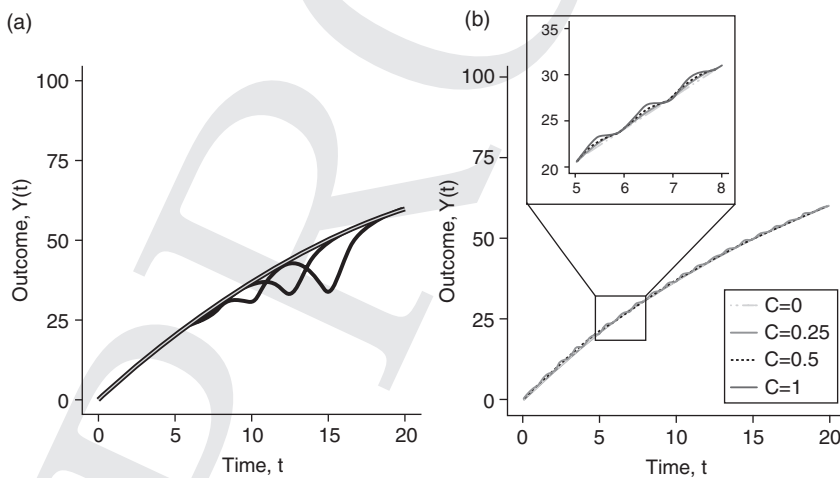


FIGURE 13.5 Depictions of two extensions of the von Bertalanffy growth model. Output of the typical model is shown as a red line. Panel A depicts outputs from the biphasic model, with variations in the extent of difference between phases, h , and timing of the phase-switch, t_h , illustrated by the four lines. Panel B depicts outputs from the seasonal model, with variations in amplitude illustrated by the three colored lines – shown most prominently within the zoomed section.

Similar in layout to the Preece-Baines model for human growth (Preece & Baines, 1978; see notational similarity with example in Grimm, Ram & Hamagami, 2011), the biphasic von Bertalanffy model provides for examination of interindividual differences in some additional features of individuals' change trajectories. Although the A_t factor was not derived from bioenergetics principles, the modification does provide for a more precise empirical description of observed growth patterns and is considered a reasonable approximation of how age-related diversion of resources toward reproductive processes affects growth. In this regard, we are immediately struck by the possibility of using the model to examine physical and behavioral changes that accompany puberty and adolescence. For example, a number of developmental theories suggest that the dual-processes governing individuals' self-regulation may be mismatched during adolescence (Steinberg, 2010). Considering that the interplay between *bottom-up processes* associated with temptations and impulses and *top-down processes* involved in deliberative evaluation can be mapped to anabolism and catabolism, and that the developmental mismatch between these two processes occurs around puberty, it seems reasonable that a biphasic model meant to accommodate reproductive investment may be useful for describing lifespan developmental changes in self-regulation. Indeed, the “dips” seen in Figure 13.5A propel the possibility of using this variant of the von Bertalanffy model for precise and parsimonious description of the psychological and social changes that manifest specifically in adolescence.

Sinusoidal growth. To accommodate some of the changes in environment that are systematically organized with respect to the Earth's rotation, the von Bertalanffy growth function has also been expanded to include a sinusoidal component. Specifically (following Somers, 1988), the seasonal von Bertalanffy model for annual cycles becomes

$$L_t = L_\infty [1 - e^{-K(t-t_0)-S_t+S_0}] \quad (13.25)$$

with

$$S_t = \left(\frac{CK}{2\pi} \right) \sin [2\pi (t - t_s)] \quad (13.26)$$

where C modulates the amplitude of the growth oscillations, the period of the cycle = 1 (see Quinn & Deriso, 1999 for generalization to cycles of any length period), t_s is a phase shift parameter that locates the cycle in time (i.e., defines the “wintering point” = $t_s + 0.5$), and L_∞ , K , and t_0 are as above. When $C = 1$ growth is completely stopped at the “wintering point” each year, and when $C = 0$, there is no seasonality to growth. Depictions of some seasonally oscillating trajectories are shown in Figure 13.5B. In particular, we see that when C gets larger the seasonality variability increases (amplitude). Different values of t_s would shift the location of the “wintering” when size is stable. Of note for interpretation, the assumption embedded in this model is that the biotic or environmental factors related to C induce sinusoidal variation in the growth rate K , which is related to seasonal

changes in the environment. As such, interindividual differences or intraindividual changes in C may describe, as noted in the earlier temperature example, factors influencing both anabolism and catabolism. Importantly, investigation of seasonal growth requires that time (e.g., age) be recorded on a fine enough scale to define the seasonal variation. For example, if seasonal growth is driven by an annual cycle, it may be necessary to locate the repeated measures with weekly or even daily precision (see Ram & Reeves, in press). In studies of human development, application of seasonal growth models to multi-year school achievement data and examination of the “forgetting” that accompanies summer holidays should be straightforward when weekly or monthly repeated measures are available.

The biphasic and sinusoidal extensions illustrate just two ways that time-varying biological and environmental factors are being incorporated into the von Bertalanffy model. On-going discussions about how to best incorporate the cost of reproduction (Minte-Vera, Maunder, Casselman, 2016) while remaining connected to the bioenergetic principles from which the models are derived (van Poorten & Walters, 2016), and about the various ways to model the plasticity of fish growth, are provoking consideration of many interesting models (Lorenzen, 2016). Notable, for instance, is consideration of how the whole framework can be recast into stochastic differential equation models that accommodate the wide variety of unknown spatially and temporally varying factors that influence growth and change at multiple time scales (Román-Román, Romero, & Torres-Ruiz, 2010; Russo, Baldi, Parisi, Magnifico, Mariani, & Cataudella, 2009). Developmental researchers will likely be able to make use of all of these extensions in their attempts to better understand humans’ physical, psychological, and social growth.

Future Directions

Our intent aim in this chapter was to explore recent and classic ecological literature on growth modeling and to develop some nascent ideas about how the models in use there may be applied to our study of human development. In that literature, we happened upon rich discussions and widespread use of the von Bertalanffy growth model, and learned that fish growth is generally conceptualized as the net result of the antagonistic processes of anabolism (the building up of body materials) and catabolism (the breaking down of body materials). As analysts looking for efficient models, we find the description of univariate change as a “dual-process” model non-parsimonious. Indeed, identification constraints (or additional assumptions) are needed to find a solution for the differential equation and actually fit the model to data. However, as developmentalists looking for representations of growth process, we find that the “dual-process” interplay between anabolism and catabolism provides the flexibility organisms need as they respond to and transact with an ever-changing environment. Indeed, the range of behaviors and trajectories we often see in our data may actually be much better represented in multivariate space than in the univariate space in which the measures were conceived and obtained.

Multivariate Growth and Measurement Models. Let us consider again the “measurement model” that was embedded in the derivation of the von Bertalanffy model for length, $W = aL^b$. In that derivation, it was mathematically convenient to take $b = 3$ in order that some exponents would cancel out in the move from a construct that exists in a 3-dimensional space (weight indicative of volume) to a construct that exists in a 1-dimensional space, length. The relation of body proportions, however, can be either simple or complex. When body proportions remain the same, the ratio of body depth to body length or head length to body length remains constant. In many species, the fish’s biomass is a cubic function of their length, $W = aL^3$, and the fishes exhibit what is called *isometric* growth. The isometric measurement model provides for straightforward data reduction and reduced data collection costs. We need only measure and model changes in fish size in one dimension, $L = \text{length}$. In contrast, when one body part grows faster or slower than another, growth is referred to as *allometric*. For fish that exhibit allometric growth, biomass generally follows a power function of length, $W = aL^b$. Usually, b is taken as invariant over time or age, but there are some species where it is necessary that the size allometry is allowed to be age-specific, $b(t)$ to accommodate, for example, substantial differences in the growth ratios among body parts for juveniles and adults (Jones, 2002).

McArdle (1988) introduced two models for multivariate growth. In a *curve-of-factors* model, where the multivariate repeated measures are modeled as indicators of a common factor that changes over time (i.e., also labelled 2nd-order growth models, Hancock, Kuo, & Lawrence, 2001). The measurement model establishing the common factor must be invariant. As in isometric growth of fish, a single change function defines how change proceeds in all dimensions. Changes in length or the common factor are directly proportional (a can be interpreted as a factor loading) to the changes in girth, or any of the indicator variables. That is, the b exponent is the same for all variables. By analogy, we can imagine growth of a cube. In the simple version of the *curve-of-factors* and isometric growth models, all the three dimensions of the cube (length, width, height) grow at the same rate, so that the cube maintains equivalence of shape over time. Changes in volume of the cube can be inferred from repeated measurement of any one dimension. Alternatively, in a *factor-of-curves* model, the repeated measures of each dimension are modeled using separate growth functions, with the parameters governing change driven by a common cause. This model is in principle like models for allometric growth, where b differs across variables (e.g., Griffiths & Sandland, 1984). The dimensions of the cube may grow at different scales. Over time, the cube with equal length sides transforms into a cuboid with unequal lengthed sides. While our and others’ implementations of the *curve-of-factors* and *factor-of-curves* models have been almost entirely focused on linear measurement models and linear change models, our hunch is that more detailed elaboration of how McArdle’s models merge with the von Bertalanffy frameworks used in biology will open up new possibilities for investigation of multivariate growth.

Time-scales and Study Design. In our scan through empirical studies of fish growth, we noted the precision with which the time-scale of growth is known and discussed. In part, this is because patterns of growth and timing of reproductive investment are extremely important in management of fisheries. Indeed, our ability to have salmon or tilapia for dinner in central Pennsylvania, or even for Jack in Hawaii, depends on knowledge of when and at what ages fish can be harvested in a profitable and sustainable way. Widespread use of the *half-life* (or doubling-time) and longevity metrics (calculated as shown in Equation 13.13) makes it very easy to locate the time-scales at which different fish grow and to plan for various interventions (e.g., temperature, food resources). In the same way that measures of *periodicity* (i.e., frequency) quantify the time-scale at which different oscillatory process manifests, measures of *half-life* quantify the time-scale at which different growth process manifest (Newell, Liu, & Meyer-Kress, 2001). We underscore this possibility because knowledge of half-life has important implications for design of longitudinal panel studies. When studying oscillatory processes, we know how to choose the measurement interval. The Nyquist-Shannon sampling theorem provides an algorithmic framework for considering how often to sample. In basic form, the sampling theorem states that the continuous signal (i.e., pattern of change) can be fully reconstructed from discrete samples obtained at a sampling frequency that is at least twice the maximum relevant bandwidth. For example, study of heart beat activity that manifests at between 0.5 and 100 Hz requires electrocardiogram sampling of at least 200 Hz. When studying incremental change processes, however, we have not been able to make such clear recommendations about when to obtain the repeated measurements. We usually wing it. When the half-life is known (or there is a clear hypothesis), however, it may be possible to make some more specific recommendations (see also Timmons & Preacher 2015). Indeed, in pharmacology, there is a body of work on optimal sampling for description of drugs' pharmacokinetic profiles (i.e., decay curves). The algorithms all seem to make use of initial knowledge of a drug's half-life (e.g., Reed, 1999; Schumacher, 1984). Although not immediately clear how this work translates to design of longitudinal panel studies of developmental processes, it is clear that quantifications or hypotheses of half-life can inform selection of sampling interval. To this end, we note that calculation of half-life requires *non-linearity*. Linear growth, which extends forever, has no discernable half-life, and the parameters used to describe straight lines are not sufficient for use in Equation 13.13. In sum, informed selection of sampling interval requires consideration of non-linear growth models, the differential forms and simulations of which facilitate construction of theoretically driven hypotheses.

Epilogue

For as long as we have known, Jack has ebulliently encouraged efforts to incorporate the use of differential equations into the analysis of behavioral change. As

he often notes (e.g., McArdle, 1988, 2009), the “lingua-franca” of differential equations provides for straightforward articulation of knowledge about change processes, highlights the simplicity and potential value of models used in other fields, and opens up a wide range of possibilities for articulating the complexities of behavioral change.

In this chapter, our engagement with the differential equation models used in the study of fish growth has expanded both the vocabulary we can now use to describe change processes and our understanding of how nonlinear growth models work. As illustrated above, the differential equations are particularly useful in connecting theory to data, because they provide for direct translation of theoretical propositions about the change process into mathematical form. Given their location in a theory to method to data research process (process-oriented theoretical model → differential equation model of change process → growth model of change outputs → empirical observations of change outputs), these models can greatly facilitate progression of knowledge about developmental and other change processes. Our general conclusion from this excursion into the fisheries literature is that many basic process models (e.g., reaction-diffusion, activator-inhibitor, anabolism-catabolism) have been worked out (also as apparent in chemistry, biology, engineering etc.). The mathematical, physical, and biological tractability of these “dual-process” models is known. Where we can see parallels between human behavior and physical and biological systems, we should make use of and/or adapt those models. Jack has always encouraged us to fish (even more strongly recently, McArdle & Ritschard, 2013 :-)) – “Just try it out and see what happens.” We are always happy we did!

Note

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298 Nilam Ram & Xiao Yang

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300 Nilam Ram & Xiao Yang

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